

Department of Information Technology
Faculty of Computing
The Islamia University of Bahawalpur

Speech Emotion Recognition Using Transfer Learning



Submitted by

1. Umyma Mohsin F22INFT1M01278
2. Muhammad Arslan Nawaz F22BINFT1M01044

Submitted to

Mr. Irfan Malik

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1. Introduction

Speech Emotion Recognition (SER) is a subfield of artificial intelligence that focuses on identifying human emotions from speech signals. It plays an important role in human-computer interaction, affective computing, virtual assistants, and mental health monitoring systems.

This project presents a deep learning-based approach for automatically classifying emotions from speech using Mel spectrogram representations and a pretrained Convolutional Neural Network (CNN), specifically VGG16.

2. Objectives

The objectives of this project are:

- To preprocess raw audio signals into structured visual representations
- To develop a deep learning model for emotion classification
- To classify speech into multiple emotional categories
- To evaluate model performance using loss and accuracy metrics

3. Dataset Description

The dataset used in this project is the **Toronto Emotional Speech Set (TESS)**.

Emotion Classes:

- Anger
- Disgust
- Fear
- Happiness
- Pleasant Surprise
- Sadness

- Neutral

The dataset contains labeled audio recordings organized into folders based on emotion types and speaker variations.

4. Methodology

4.1 Audio Preprocessing

Each audio file is processed using the Librosa library:

- Audio is loaded at a sampling rate of **16,000 Hz**
- Converted into **Mel spectrograms** with 128 Mel bands
- Spectrograms are transformed into **decibel (dB) scale**
- Each spectrogram is resized to a fixed shape of **128×128**
- Padding or truncation is applied to ensure uniform input size
- Spectrograms are replicated into **3 channels** to match CNN input requirements

This transforms audio signals into image-like representations suitable for CNN processing.

4.2 Dataset Preparation

A custom PyTorch dataset class (`EmotionDataset`) is implemented to:

- Load audio files dynamically from directory structure
- Assign numeric labels to each emotion class
- Convert audio into Mel spectrogram tensors
- Return processed data for training

The dataset is split as:

- 70% Training Set
- 15% Validation Set
- 15% Test Set

4.3 Model Architecture

The model is based on **VGG16 pretrained on ImageNet**:

- Convolutional layers are used as a feature extractor
- All pretrained weights are frozen to preserve learned features
- The final classification layer is replaced with a custom fully connected layer

This enables transfer learning, allowing the model to adapt image-based features to audio spectrogram data.

4.4 Training Configuration

- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Learning Rate: 0.0001
- Batch Size: 32
- Epochs: 15

5. Training Process

During each training epoch:

- Input spectrograms are passed through the model
- Predictions are generated as class probabilities
- Loss is calculated using CrossEntropyLoss
- Gradients are computed using backpropagation
- Model weights are updated using the Adam optimizer

Training performance is tracked using:

- Training Loss
- Training Accuracy

6. Model Performance (Accuracy and Loss Analysis)

6.1 Loss Function

CrossEntropyLoss is used for multi-class classification. It penalizes incorrect predictions and encourages correct class probability distributions.

6.2 Accuracy Definition

- [
 \text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}}
]
- Accuracy is computed using predicted class (argmax) compared to true labels.
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6.3 Training Results

- The following table summarizes training performance across 15 epochs:

Epoch	Train Loss	Train Accuracy
1	1.0950	0.7200
2	0.7898	0.7871
3	0.6159	0.8171
4	0.5873	0.8300
5	0.5550	0.8514
6	0.4292	0.8571
7	0.3786	0.8914
8	0.3080	0.8986
9	0.2891	0.9086
10	0.2632	0.9214
11	0.2405	0.9271
12	0.3421	0.9029
13	0.2580	0.9143
14	0.2795	0.9186
15	0.2114	0.9386

6.4 Observations

- Loss decreases significantly from 1.0950 to 0.2114
- Accuracy improves from 72% to 93.86%
- Minor fluctuations are observed due to batch variability

- Overall trend indicates stable convergence and effective learning

7. Model Saving

The trained model is saved using PyTorch:

- Stored as .pth file in Google Drive
- Contains model state dictionary for future inference

8. Applications

Speech Emotion Recognition systems can be used in:

- Virtual assistants with emotional awareness
- Customer service sentiment analysis
- Mental health monitoring systems
- Human-computer interaction systems
- Interactive entertainment and gaming

9. Limitations

- Dataset size is limited to controlled speech samples
- Background noise is not explicitly handled
- Model does not capture temporal dependencies in speech
- Transfer learning uses image-based features for audio data

10. Future Improvements

- Fine-tuning pretrained VGG16 layers instead of freezing them
- Adding data augmentation (noise injection, pitch shifting)
- Using sequential models like CNN-LSTM for temporal learning
- Exploring transformer-based audio models (e.g., Wav2Vec)
- Deploying as a real-time emotion recognition system

11. Conclusion

This project demonstrates a deep learning approach to Speech Emotion Recognition using Mel spectrograms and transfer learning with VGG16. The model successfully classifies emotions from speech with improving accuracy and decreasing loss over training epochs.

The results show that spectrogram-based CNN models are effective for audio emotion classification tasks, with potential for real-world deployment in interactive and assistive technologies.

12. Tools and Technologies

- Python
- PyTorch
- Librosa
- NumPy
- Matplotlib
- Google Colab

13. References

- Librosa Documentation
- PyTorch Documentation
- Toronto Emotional Speech Set (TESS) Dataset

